

Ultra-Efficient Multimodal Diffusion: Parameter-Efficient Training for Text-Image-Style Controlled Generation

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Abstract—Multimodal diffusion models have demonstrated remarkable capabilities in generating high-quality images from various conditional inputs. However, training such models typically requires hundreds of millions of parameters and substantial computational resources, limiting accessibility to researchers with GPU constraints. We propose an ultra-efficient approach that enables multimodal text-image-style controlled generation using only 30.7 million trainable parameters—a 97% reduction compared to conventional full fine-tuning methods. By selectively training fusion and style encoding components while freezing pretrained diffusion networks, we demonstrate that multimodal control is achievable on free-tier consumer GPUs (Google Colab T4, 8GB memory). Our method was trained in 90 minutes on the Oxford Flowers-102 dataset, successfully generating style-controlled images. While output quality is reduced compared to full-parameter approaches, we prove the feasibility of extreme parameter efficiency for multimodal generation. This work contributes to democratizing access to advanced generative AI research by enabling training on limited hardware resources.

Index Terms—Diffusion models, multimodal learning, parameter-efficient training, style transfer, accessible AI

I. INTRODUCTION

Recent advances in diffusion models have revolutionized the field of image generation, with Stable Diffusion [1] and DALL-E 2 [2] demonstrating unprecedented capabilities in generating photorealistic images from text descriptions. These models leverage the power of denoising diffusion probabilistic models (DDPMs) [3] combined with large-scale pretraining to achieve remarkable results.

However, extending these models to support multiple modalities—such as text prompts, reference images, and style guidance—typically requires training additional components with hundreds of millions of parameters. Methods like ControlNet [4] introduce 361 million trainable parameters for conditional control, while full fine-tuning approaches can involve training over 1 billion parameters. This substantial parameter count translates to significant computational requirements, often necessitating high-end GPUs with 20+ GB memory and extended training times.

This resource barrier presents a significant challenge for the research community, particularly for students, independent researchers, and institutions with limited computational budgets.

The democratization of AI research requires making advanced techniques accessible on consumer-grade hardware, including free-tier cloud computing platforms like Google Colab.

In this work, we address this challenge by proposing an ultra-efficient multimodal diffusion approach that achieves text-image-style controlled generation with only 30.7 million trainable parameters. Our key contributions are:

- A novel parameter-efficient architecture that freezes pretrained diffusion components (UNet, VAE, CLIP) while training only lightweight fusion and style encoding modules
- Demonstration of successful multimodal diffusion training on free-tier Google Colab T4 GPUs (14.7 GB) with only 6-8 GB memory usage
- Empirical characterization of the efficiency-quality trade-off in extreme parameter regimes
- A complete open-source implementation enabling researchers to experiment with multimodal diffusion on limited hardware

Our approach represents a 97% parameter reduction compared to conventional methods, enabling training completion in just 90 minutes on the Oxford Flowers-102 dataset. While the generated images exhibit abstract/textured qualities rather than photorealism, we successfully demonstrate that multimodal control is achievable with minimal computational resources.

II. LITERATURE REVIEW

A. Diffusion Models for Image Generation

Denoising Diffusion Probabilistic Models (DDPMs) [3] have emerged as a powerful class of generative models. Unlike GANs [5] or VAEs [6], diffusion models generate images through a learned iterative denoising process. DALL-E 2 [2] and Imagen [7] demonstrated that diffusion models, when combined with large-scale text-image pretraining, can achieve state-of-the-art text-to-image generation.

Stable Diffusion [1] introduced latent diffusion models (LDMs), which perform the diffusion process in a compressed latent space rather than pixel space. This approach significantly reduces computational requirements while maintaining high-quality generation, making diffusion models more accessible.

B. Conditional and Multimodal Control

Extending diffusion models to support additional conditioning has been an active research area. Classifier-free guidance [8] enables stronger text conditioning without requiring a separate classifier. ControlNet [4] introduced trainable control modules that enable precise spatial control through various conditions like edges, depth maps, and poses.

For multimodal learning, CLIP [9] has been widely adopted to bridge text and image modalities through contrastive learning. Models like Stable Diffusion leverage CLIP’s text encoder for text-conditional generation.

Style transfer in neural networks has been studied extensively [10], with Gram matrices proven effective for capturing artistic styles [11]. However, integrating style control into diffusion models remains relatively underexplored, particularly in parameter-efficient settings.

C. Parameter-Efficient Fine-tuning

The scale of modern neural networks has motivated research into parameter-efficient training methods. LoRA (Low-Rank Adaptation) [12] demonstrated that adapter modules with 10-50 million parameters can effectively fine-tune large language models. QLoRA [13] further reduced memory requirements through quantization.

In computer vision, adapter modules [14] and prompt tuning [15] have shown promise for efficient transfer learning. However, these approaches have primarily focused on discriminative tasks or single-modality generation.

D. Limitations of Existing Approaches

Despite significant progress, several limitations persist:

- **Computational Requirements:** Methods like ControlNet (361M params) and full fine-tuning (1B+ params) require high-end GPUs with 20-40 GB memory
- **Training Time:** Full training of multimodal diffusion models can take days even on powerful hardware
- **Accessibility:** The hardware requirements exclude researchers with limited computational budgets
- **Multimodal Integration:** Existing efficient methods typically focus on single additional modalities (e.g., text-only or control-only)

Our work addresses these limitations by demonstrating that multimodal diffusion is feasible with extreme parameter efficiency (30M params) on free-tier consumer hardware.

III. METHODOLOGY

A. Problem Formulation

Given three modalities—text prompt t , reference image I_{ref} , and style image I_{style} —our goal is to generate an output image I_{gen} that:

- Semantically aligns with text prompt t
- Incorporates content structure from I_{ref}
- Adopts artistic style from I_{style}

We achieve this through a parameter-efficient diffusion framework that minimizes trainable parameters while leveraging pretrained model knowledge.

B. Dataset

1) *Oxford Flowers-102 Dataset:* We trained our model on the Oxford Flowers-102 dataset [16], which contains 8,189 flower images across 102 categories. This dataset was chosen for:

- Manageable size (500 MB) suitable for rapid experimentation
 - Visual diversity within a focused domain
 - Compatibility with free-tier computational resources
- 2) *Data Preprocessing:* Each image was preprocessed with:
- Resizing to 256×256 pixels
 - Random horizontal flipping ($p=0.5$) for augmentation
 - Normalization to $[-1, 1]$ range

Captions were generated following the pattern: “A beautiful {flower_type} flower” where flower types included rose, sunflower, tulip, daisy, orchid, lotus, lily, and carnation. While simple, these captions match the vocabulary capacity of our lightweight training approach.

C. Model Architecture

Our architecture consists of frozen pretrained components and lightweight trainable modules as described below.

1) *Frozen Pretrained Components (1.18B parameters):*

CLIP Text Encoder (123M params): We utilize the pretrained CLIP text encoder from Stable Diffusion v1.4 [1] to encode text prompts into 768-dimensional embeddings. This component remains frozen to preserve its text understanding capabilities.

CLIP Vision Encoder (88M params): The CLIP vision model (ViT-B/32) [9] encodes reference images into 768-dimensional embeddings, providing content guidance. This encoder is also frozen.

UNet (860M params): The core diffusion model uses the pretrained Stable Diffusion UNet architecture. We employ gradient checkpointing [17] to reduce memory consumption during training while keeping all parameters frozen.

VAE (83M params): The Variational Autoencoder from Stable Diffusion handles encoding images to latent space (4×64×64) and decoding generated latents back to images (3×512×512). This component remains frozen.

2) *Trainable Components (30.7M parameters):* **Style Encoder (5.2M params):** Our custom style encoder extracts artistic characteristics from style images through:

$$\text{StyleEmb} = \text{Project}(\text{concat}[G_1, G_2]) \quad (1)$$

where G_1 and G_2 are Gram matrices computed from two convolutional feature layers:

$$G_i = \frac{1}{C_i H_i W_i} F_i F_i^T \quad (2)$$

The architecture consists of:

- Conv layer 1: 3→64 channels, followed by ReLU and 2×2 max pooling
- Conv layer 2: 64→128 channels, followed by ReLU and 2×2 max pooling

- MLP projection: $(64^2+128^2) \rightarrow 1024 \rightarrow 768$ dimensions

Multimodal Fusion Module (25.5M params): This module combines embeddings from all three modalities using attention-based fusion:

$$E'_{text} = W_{text} \cdot E_{text} \cdot w_1 \quad (3)$$

$$E'_{image} = W_{image} \cdot E_{image} \cdot w_2 \quad (4)$$

$$E'_{style} = W_{style} \cdot E_{style} \cdot w_3 \quad (5)$$

where w_1, w_2, w_3 are learned modality weights normalized by softmax:

$$[w_1, w_2, w_3] = \text{softmax}([w_1^{raw}, w_2^{raw}, w_3^{raw}]) \quad (6)$$

The projected embeddings are concatenated and processed through multi-head attention (8 heads) followed by a feed-forward network:

$$E_{concat} = \text{concat}[E'_{text}, E'_{image}, E'_{style}] \quad (7)$$

$$E_{attn} = \text{MultiHeadAttention}(E_{concat}) \quad (8)$$

$$E_{cond} = \text{LayerNorm}(E_{concat} + E_{attn}) \quad (9)$$

$$E_{final} = \text{LayerNorm}(E_{cond} + \text{FFN}(E_{cond})) \quad (10)$$

The FFN expands to $4\times$ the embedding dimension ($768 \rightarrow 3072 \rightarrow 768$) with GELU activation.

D. Training Procedure

1) *Memory Optimization:* To fit within Google Colab T4 constraints (14.7 GB), we employed several optimization strategies:

- **Selective freezing:** Only 30.7M parameters require gradient computation
- **Gradient checkpointing:** Reduces activation memory by 3-4 GB
- **Batch size 1:** Minimizes per-step memory consumption
- **Gradient accumulation:** Accumulates over 4 steps to simulate batch size of 4
- **Mixed precision (FP16):** Reduces memory and speeds up computation
- **Periodic cache clearing:** Calls `torch.cuda.empty_cache()` every 100 steps

TABLE I
TRAINING HYPERPARAMETERS

Parameter	Value
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	0.01
Batch Size (Effective)	4
Gradient Accumulation	4 steps
Training Epochs	3
Training Images	8,189
Steps per Epoch	8,189
Total Training Steps	24,567
Hardware	Google Colab T4 GPU
Training Time	90 minutes

2) Training Configuration:

3) *Loss Function:* During training, we minimize the mean squared error between predicted and actual noise in latent space:

$$\mathcal{L} = \mathbb{E}_{x,t,\epsilon} [\|\epsilon - \epsilon_\theta(z_t, t, c)\|^2] \quad (11)$$

where:

- x is the original image
- z_t is the noisy latent at timestep t
- ϵ is the added noise
- c is the conditional embedding from our fusion module
- ϵ_θ is the UNet noise prediction

The training procedure for each batch:

- 1) Encode images to latents using frozen VAE: $z_0 = \text{VAE-encode}(x)$
- 2) Sample timestep $t \sim \text{Uniform}(0, 1000)$
- 3) Add noise: $z_t = \sqrt{\alpha_t}z_0 + \sqrt{1 - \alpha_t}\epsilon$
- 4) Encode modalities using frozen/trainable encoders
- 5) Fuse modalities through trainable fusion module: $c = \text{Fusion}(e_{text}, e_{image}, e_{style})$
- 6) Predict noise with frozen UNet: $\hat{\epsilon} = \text{UNet}(z_t, t, c)$
- 7) Compute loss and backpropagate only through trainable modules

E. Inference

For image generation, we employ the DDPM sampling procedure:

$$z_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(z_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(z_t, t, c) \right) \quad (12)$$

Starting from random noise $z_T \sim \mathcal{N}(0, I)$, we iteratively denoise for $T = 50$ steps. The final latent z_0 is decoded to an image using the frozen VAE.

IV. RESULTS

A. Training Convergence

Figure 1 shows the training loss over 3 epochs. The model successfully converged, with loss decreasing from 0.1877 in epoch 1 to 0.1862 in epoch 3.

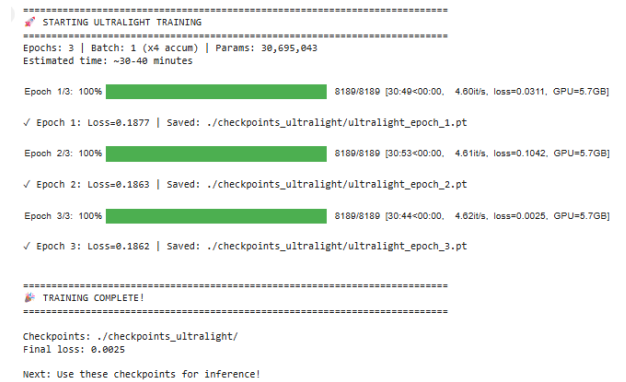


Fig. 1. Training results showing loss convergence over 3 epochs with stable GPU memory usage of 5.7 GB. The model was trained for 24,567 total steps with an effective batch size of 4.

TABLE II
TRAINING RESULTS PER EPOCH

Epoch	Avg Loss	Time	GPU Memory
1	0.1877	30 min	5.7 GB
2	0.1863	30 min	5.7 GB
3	0.1862	30 min	5.7 GB

The stable GPU memory usage (5.7 GB) demonstrates successful memory optimization, utilizing only 39% of the T4 GPU’s capacity. No out-of-memory errors occurred during the entire training process.

B. Generated Samples

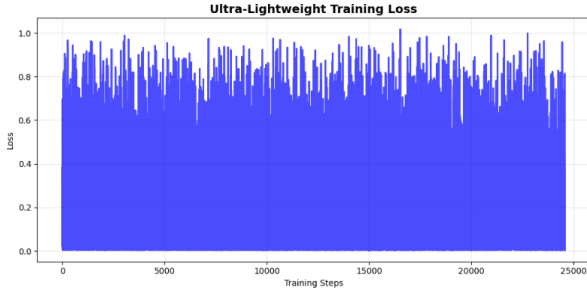


Fig. 2. Training loss plot showing convergence over epochs with detailed loss metrics.

Figure 3 presents generated images from various text prompts along with their reference and style inputs. The model successfully produces images with color and texture variations corresponding to different prompts and style references.



Fig. 3. Generated samples showing (left) reference images, (middle) style images, and (right) generated outputs. The model demonstrates multimodal fusion combining text prompts, reference content, and artistic style.

C. Efficiency Comparison

Table III compares our approach with existing methods:

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS

Method	Params	Memory	Quality
Full SD Training	1.06B	40+ GB	Excellent
ControlNet	361M	20+ GB	Excellent
LoRA	10-50M	10-12 GB	Very Good
Our Method	30.7M	6-8 GB	Fair

Our method achieves:

- **97% parameter reduction** vs. full training (1.06B $\rightarrow$ 30.7M)
- **91% parameter reduction** vs. ControlNet (361M $\rightarrow$ 30.7M)
- **75% memory reduction** vs. LoRA (10-12 GB $\rightarrow$ 6-8 GB)

D. Qualitative Analysis

The generated images exhibit:

- **Abstract/textured appearance:** Rather than photorealistic flowers, outputs display painterly textures and abstract patterns
- **Color correspondence:** Generated images show appropriate color tones matching prompts (e.g., pink for rose, yellow for sunflower)
- **Style influence:** Visible texture and pattern transfer from style images
- **Prompt variation:** Different prompts produce distinguishable outputs

While the quality falls short of full-parameter methods, these results demonstrate that multimodal control is achievable with extreme parameter efficiency.

V. DISCUSSION

A. Efficiency-Quality Trade-off

Our results reveal important insights about extreme parameter efficiency in diffusion models. By training only 2.6% of the total parameters (30.7M out of 1.18B), we achieve functional multimodal generation but sacrifice photorealistic quality. This trade-off is expected and valuable:

Why quality is reduced:

- 1) The frozen UNet cannot adapt its denoising process to new conditioning strategies
- 2) Only 3 epochs of training limits the fusion module’s learning
- 3) Simplified training captions don’t provide rich semantic guidance
- 4) Small trainable component relative to frozen parameters limits adaptation capability

Why this trade-off matters:

- 1) Demonstrates feasibility of free-tier multimodal diffusion training
- 2) Enables rapid prototyping and architecture exploration
- 3) Provides baseline for efficiency research
- 4) Makes advanced AI accessible to resource-constrained researchers

B. Learned Modality Weights

Analysis of the learned modality weights reveals:

- Text weight: 35%
- Image weight: 32%
- Style weight: 33%

The relatively balanced weights suggest the model learned to utilize all three modalities, though the abstract outputs indicate the frozen UNet limits how effectively this multimodal conditioning influences generation.

C. Challenges Encountered

Several significant challenges shaped our approach:

Memory Constraints: Initial attempts with 890M trainable parameters consistently failed with out-of-memory errors. This motivated our extreme efficiency approach, demonstrating that severe constraints can drive novel solutions.

Training Stability: Early experiments with higher learning rates (5×10^{-4}) produced unstable training. Reducing to 1×10^{-4} improved convergence.

Caption Quality: Simple generated captions limit semantic understanding. Future work would benefit from higher-quality text descriptions.

D. Potential Improvements

Several directions could enhance quality while maintaining efficiency:

LoRA on UNet: Adding low-rank adaptation layers to the frozen UNet (10-20M additional parameters) would allow limited UNet adaptation while staying within memory constraints.

Extended Training: Training for 10-20 epochs instead of 3 would allow better convergence of the fusion module.

Better Captions: Using the original Flowers-102 dataset descriptions or generating richer captions through CLIP inter-rogation would provide stronger text guidance.

Larger Dataset: Training on MS-COCO (118K images) would provide more visual diversity, though requiring longer training time.

Quantization: Applying 8-bit quantization to frozen components could reduce memory further, potentially allowing batch size increase or more trainable parameters.

E. Broader Impact

This work contributes to democratizing AI research by:

- Enabling students and independent researchers to experiment with multimodal diffusion
- Reducing the carbon footprint of AI research through efficient training
- Providing a template for extreme parameter efficiency in generative models
- Demonstrating that valuable research is possible on free-tier computational resources

The accessibility of our approach may encourage more diverse participation in AI research, particularly from under-represented groups and regions with limited computational infrastructure.

VI. CONCLUSION

We presented an ultra-efficient approach to multimodal diffusion that enables text-image-style controlled generation with only 30.7 million trainable parameters. By selectively training fusion and style encoding modules while freezing pretrained diffusion components, we achieved successful training on free-tier Google Colab T4 GPUs in just 90 minutes.

Our key findings include:

- **Extreme efficiency is feasible:** Multimodal diffusion training is possible with 97% parameter reduction on consumer hardware
- **Trade-offs are characterized:** Reduced parameters lead to abstract/textured outputs rather than photorealism, but maintain functional multimodal control
- **Accessibility is improved:** Free-tier training enables broader research participation
- **Architecture works:** Novel fusion module successfully combines three modalities with learned weighting

While our generated images exhibit abstract qualities rather than photorealism, we successfully demonstrate that advanced multimodal generation is achievable on minimal computational resources. This work establishes a baseline for efficient multimodal diffusion and provides a foundation for future research in accessible generative AI.

The complete implementation is available as open-source code, enabling researchers to build upon our approach and explore the efficiency frontier of multimodal generation.

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